GRaD-Nav Real4Real audit

Native rendering and dynamics coupled Policy and fidelity remain blocked

One existing A6000 host. No training. Instance remains running.

Decision: study native rendering and dynamics. Defer calibrated Real4Real and policy claims.

Real4Real evidence still needed

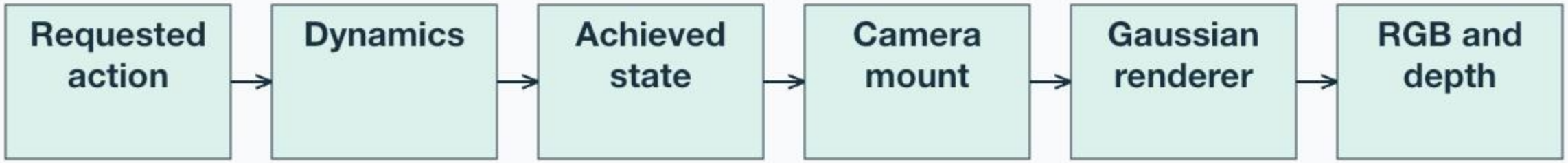
Outdoor residential, suburban and campus scenes below 100 m

Held-out horizontal and vertical camera trajectories

Capture at 15 m and rendering at 50 m remains a research question

Sparse capture coverage and sparse physical content need separate tests

Action, state and observation path



B: measured

Mounted pose checked

A/D: rendered

C needs the original actor, normalization, VAE and trained visual observation components.

Released scene assets recovered

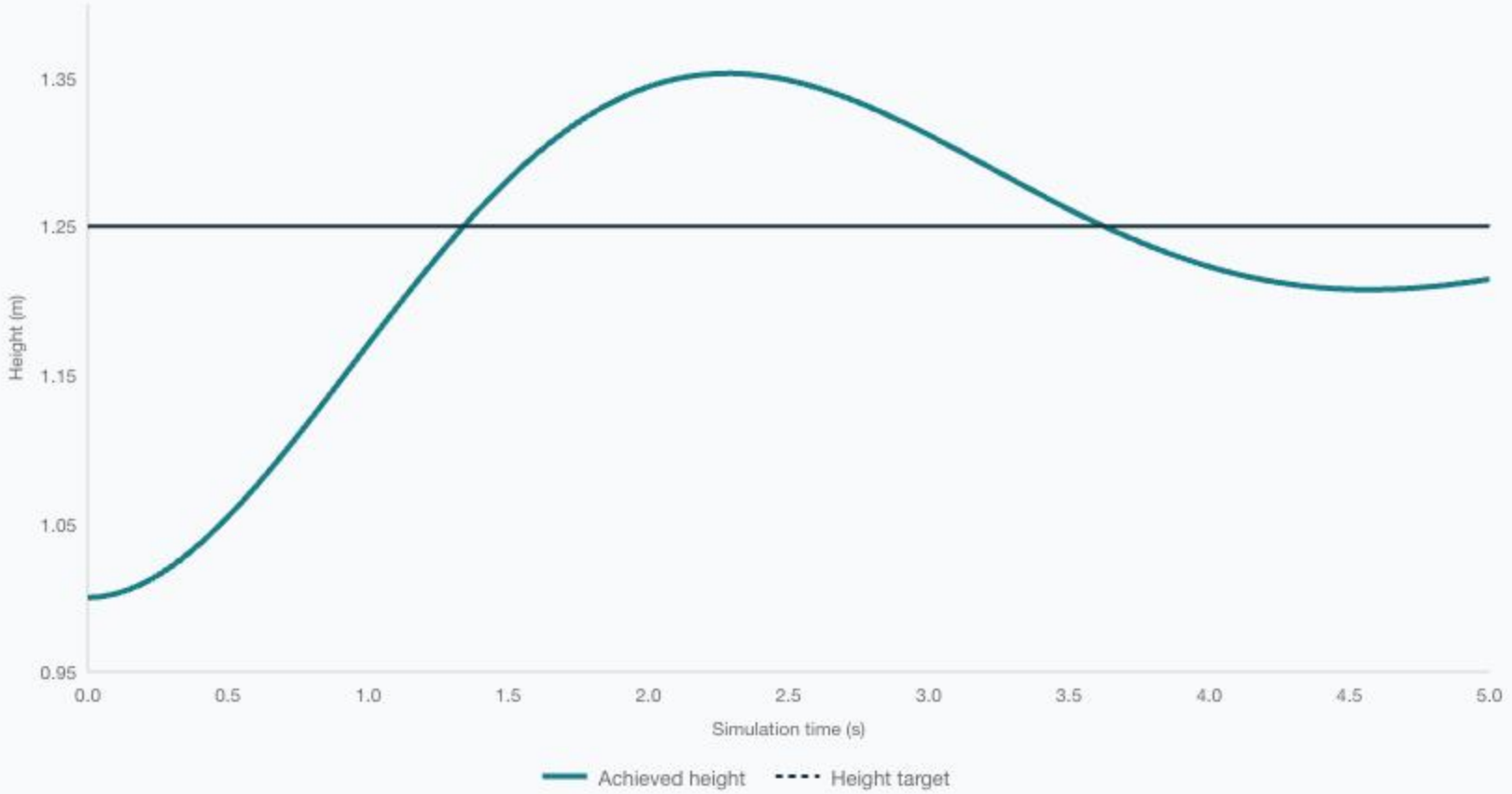
README file IDs returned 404. The author folder supplied replacement IDs.

Gaussian archive 5.049 GB. Full expanded size 5.827 GB.

Point-cloud archive 0.294 GB. Selected cloud 65.9 MB.

Selected gate_mid inputs: 1.140 GB across 1,213 locally verified files.

B feedback response



Final error
0.0356 m

Overshoot
0.1030 m

First 0.02 m band entry
1.25 s

The error band is an authored diagnostic. First entry does not establish settling or stable hover.

B scope and measured behavior

Force balance: 20 × 0.05 s. Height 1.25 m, drift 0.0 at float32 precision.

Feedback: kp 2, kd 0.8. Initial perturbation −0.25 m.

One second: drift +0.1706 m, remaining error 0.0794 m, no overshoot.

Five seconds: drift +0.2144 m, error 0.0356 m, overshoot 0.1030 m.

No saturation. Dynamics median about 14.2 ms, with GPU synchronization. Rendering timing is reported separately; policy latency remains blocked.

Numerical camera checks

Camera helper uses xyzw. Dynamics uses wxyz. Conflicting comments are recorded.

Identity, 90° yaw and translation composition checks passed.

Active helper applies no dataparser normalization. Scale/alignment needs scene validation.

Rounded mount orthonormality residual: 0.00042425.

These are algebraic checks. Direct/author pixel parity is exact. Native D view is upright. Capture calibration remains unresolved.

Compiler selection repaired

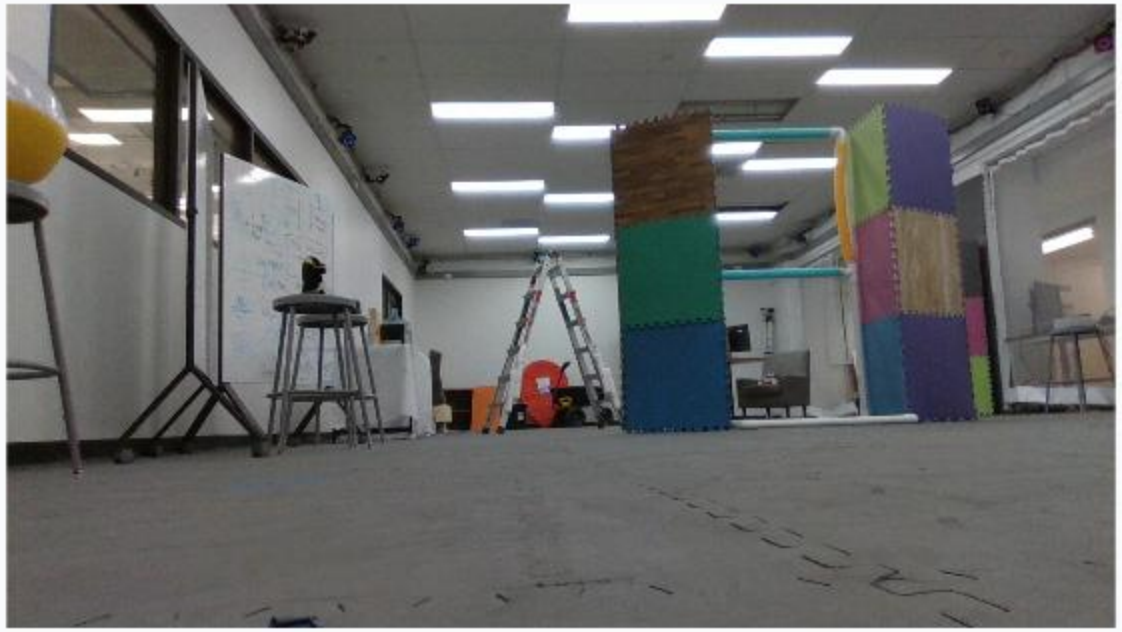
Default GCC 12 cannot resolve cc1plus.
The minimal CUDA control reproduces the exact failure.

GCC 11 C++ and CUDA tests compile and run successfully.

Emitted gsplat command: -ccbin /usr/bin/gcc-11.
CXX: /usr/bin/g++-11. Binary linked and rendered.

One repair attempt. No package, driver or alternative changes.
Debugging stopped after 12.12 minutes; second attempt unused.

A: equal pixels, unresolved capture view



Associated author capture



Frozen A output

All five direct/author RGB, depth and camera differences: 0. COLMAP/source poses agree exactly. Capture-to-checkpoint alignment remains unresolved.

A rendering latency and depth

Direct render: median 2.765 ms; p95 2.802 ms.

Author render and pose/stack: median 3.426 ms; p95 3.490 ms.

Raw RGB/depth + CPU transfer: median 3.500 ms; p95 3.595 ms.

Peak Torch allocated 174.2 MB; driver/context totals sampled separately.

RGB+ED is expected camera-axis depth. No-hit diagnostic yields alpha 0 and depth 0. Learned-feature/policy timing remains blocked.

D: achieved state determines the image



First frame: upright native gate view



Final frame: follows achieved height

Height 1.15 → 1.21829; controller kp 2, kd 0.8.
Coordinate coupling works. Metric flight, task success and outdoor fidelity remain blocked.

Native policy and collision limits

C blocked: scene checkpoints do not supply the original six learned components.

The native body-rate termination check reads quaternion fields 10:13.

Obstacle handling uses forward-FOV nearest-point distance and reward.

Rear and coincident synthetic points returned infinity. No body-volume contact test.

Outdoor fidelity remains untested

No PSNR, SSIM or LPIPS result. No validated held-out reconstruction split.

No altitude-change, sparse-capture, vegetation or moving-object experiment.

Indoor/gate paper results motivate study. Outdoor and racing were not executed.

Other candidate scenes, controllers and hardware differ. No matched ranking.

Decision, preservation and next evidence

Native rendering and dynamics work as research components. Defer calibrated Real4Real and policy reproduction.

All 16 run directories and 1,213 scientific files verified locally.
Logs, configs, route records and compiler failures are preserved.

Next evidence: author-confirmed scene/capture registration, metric scale, original learned components and trusted held-out real capture.

Provenance appendix

Upstream baseline: d9b00e16107338e49071c65f44582c013e522633

Current source inspected: 95218abe9a5d355519cdaab78d915365303066fe

Host: Ubuntu 22.04.5, RTX A6000, 6 vCPU, 47 GiB RAM

Rendering stack: Torch 2.1.2/cu118, NS 1.1.3, gsplat 1.0.0.

Full-instance charges unknown. $0.54/hour is an unconfirmed GPU-only quote. No termination or cleanup occurred.